

Guidelines for statement extraction

In this document, we define what an atomic statement extracted from a linguistic article is, and establish the annotation rules. The goal of this guideline is to answer the questions:

- Is the statement correct? / Does it correspond to the context?
- Is it independent of the context?
- Is it sufficiently atomic?

The annotation process consists of:

1. Assign to each statement one of the two possible tags: “Yes” or “Problem”.
 - a. If marked as “Problem”, provide a correct version of the statement
2. If there are statements in the context that haven't been extracted, add them in a new line.

1. Definition of statement

A statement is a sentence that,

1. Can be directly inferred from the source text and has to be a literal interpretation of the source text
 - Source text: “ $A \rightarrow B$; $B \rightarrow C$ ”, only the two literally present statements should be extracted and not “ $A \rightarrow C$ ”.
2. It's independently understandable and interpretable on its own:
 - All references to other parts of the text, such as pronouns and definite articles, have to be replaced by the actual referee.
 - Thus, there are no references, such as “they”, “these models”, “the previous results”, etc.

3. It's atomic. An atomic statement is a sentence that can not be split into a conjunction of statements without losing its meaning. i.e. each coordination has to be split into separate statements (with the exception of lists of references, see below).

- If it is ambiguous whether a statement is atomic or not, it should be considered atomic.

- Rules on implication:

- Implications usually come in the form of "A and $A \rightarrow B$ ".

Example:

Given the source text: "Prepositions also act as cues to retrieve the PP correlate. **Hence**, P-omission in sluices comes with a processing cost." ($A \rightarrow B$)

In this case, we do not want B, i.e. "Omitting the preposition (P-omission) in sluicing constructions incurs a processing cost for comprehenders", as an atomic statement.

But we want A ("Prepositions also act as cues to retrieve the PP correlate."), and we want $A \rightarrow B$ as a separate statement ("Prepositions acting as cues to retrieve the PP correlate imply that P-omission in sluices comes with a processing cost.")

- Pure implications " $A \rightarrow B$ " are considered atomic; A and B should not be given as separate atomic claims.

- Rules on relative pronouns:

1. For non-restrictive "which": e.g., for "French, which is an Indo-European language, ...", the statement "French is an Indo-European language" should be extracted.

2. If it is uncertain whether the "which" is non-restrictive, it should not be split into two statements.

Ambiguous cases:

1. If the author owns a statement (e.g., “This paper argues that Y is Z”, do not include “the author/article/paper argues/assumes/states/proposes that...” or similar phrases in the extracted statement. The statement should be simply “Y is Z”.

1.1. Citations and references specifications

Information about how references should be handled in the extracted statements:

- All the statements that contain a citation, for example, “X (2024) uses Y to prove Z”, should be split if possible into “X uses Y to prove Z” and a potential sentence “Y can be used to prove Z”
- statement as a citation + statement in general
- The statement must contain the reference if the author of the source text attributes the statement to another paper.
 - In cases of doubt about the scope of a citation, we assume that the author attributes only the text preceding the in-text citation. If there is ambiguity on the scope of a citation, it should always be considered the smallest scope.

For example, for the context “Talmeri is a morphologically ergative language in which agentive intransitives take an ergative subject and kar “have,” while patientive intransitives take an absolutive subject and sen “be” (X 1998).”, we consider that the statement “Talmeri is a morphologically ergative language” is out of the scope of the citation because it can be considered as a general claim about the language.

- The scope of a citation may be in doubt in the following cases:
 1. Citation applies to text left or right: (e.g., “Under direct interpretation approaches to ellipsis (Ginzburg & Sag 2000), an interrogative sluice is

not derived from a full interrogative counterpart”). In such cases, we assume that the in-text citation is only about the text preceding it.

2. Citation applies to text how far to the left: When the citation is preceded by a long sentence with many clauses, and it is unclear which ones are covered, the citation is assumed to cover only the closest clause.
- In the case of lists of references to whom the author conjointly attributes the same claim, they should all appear in the same statement.
 - If the cited text is between quotes, it should appear between quotes exactly the same in the statement. It is not necessary to split it into more atomic claims.
 - Given the source structure “X (2024) followed Y (2023) in their analysis of sluicing ...” (citation). Expected statements:
 1. “Y (2023) analyses sluicing...”
 2. “X (2024) uses the same analysis methodology as Y (2023) for sluicing ...”

2. Examples of annotation

2.1. “YES” examples

The sentence is a statement.

Example:

Context: Through the analysis of a large (written) corpus, we also show that P-omission in French sluices is the preferred option, as in English (Nykiel 2017), and identify the factors favouring it.

Correct statement: In a large written corpus, P-omission in French sluices is the preferred option.

2.2. “PROBLEM” examples

The sentence breaks at least one of the conditions to be a statement; thus, it's badly extracted.

Example:

Context: A male author/researcher claims that languages that do not allow preposition stranding ban preposition omission under sluicing.

Problematic statement: Merchant (2001) claims that non-P-stranding languages ban P-omission under sluicing.